

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

UNIT AND

Dimension

LECTURE-3

PHYSICS

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Topics To Covered

1

RIVISION

2

DIMENSION

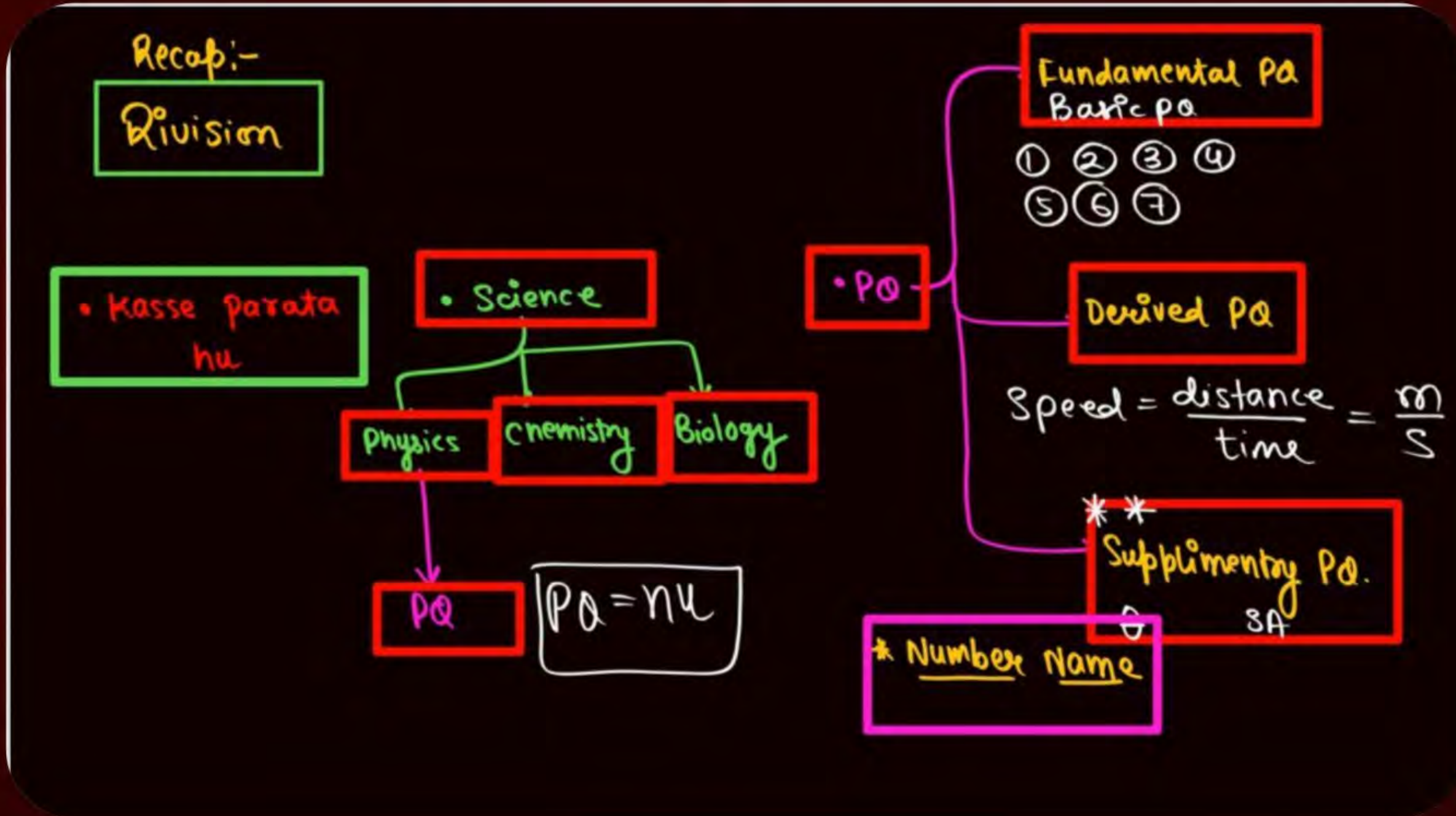
3

APPLICATION OF DIMENSION

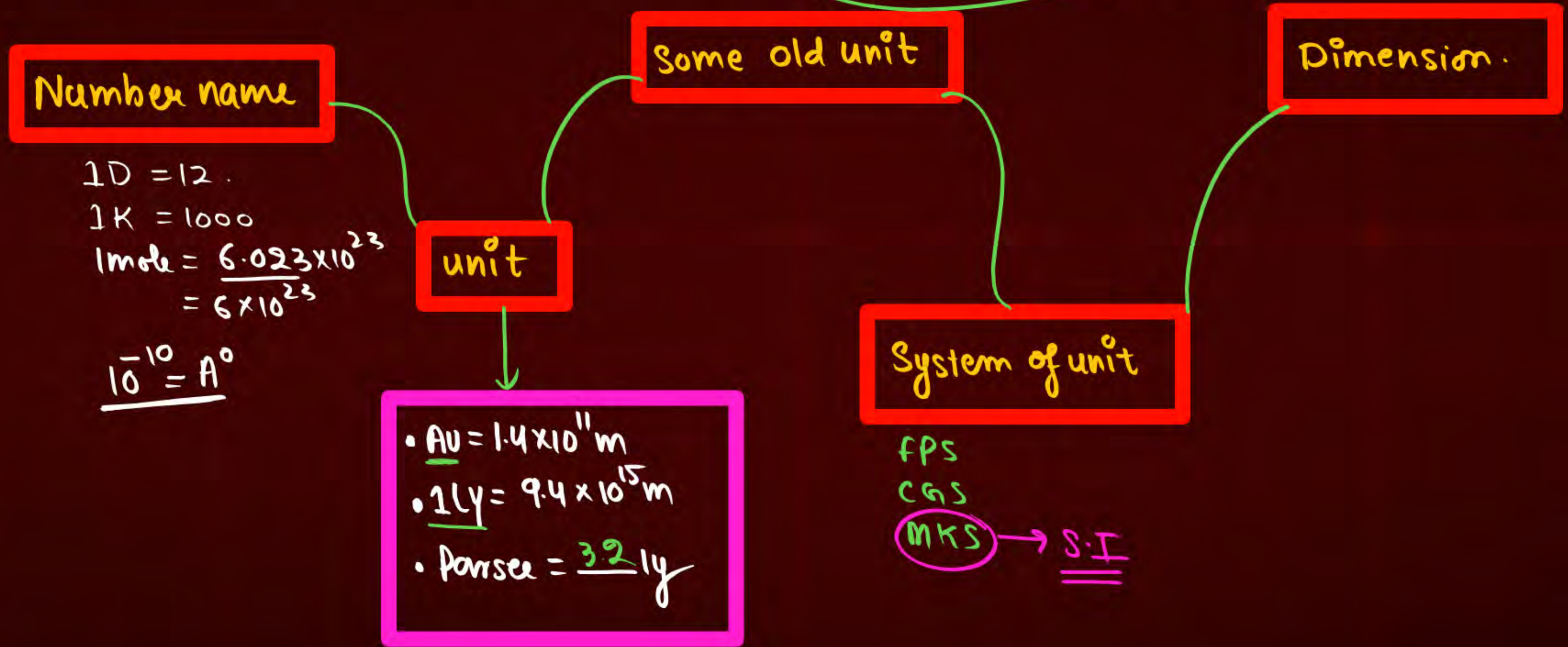
4

a) TO FIND OUT THE DIMENSIONAL FORMULA OF PQ

Recap From 1 Lec.



Recap From 2. Lec.



Dimension

→ Newton chacha
• old unit

The dimensions are the Powers raised to base Quantity to determine the relationship b/w base pq and derived pq

- ① Mass → M (kg)
- ② Length → L (m)
- ③ Time → T (Sec)
- ④ Current → A Ampere
- ⑤ Temperature → K / θ
- ⑥ Luminous intensity → Cd
- ⑦ No. of Substance → → mol

• Speed = $\frac{\text{distance}}{\text{time}}$ $\frac{S}{2}$

Speed = $\frac{L}{T}$

Speed = $L^{-1} T^{-1}$ = $L T^{-1}$

Dimensional equation

Dimensional formula

Dimension



Note :- ✓ Jab bi apko **Dimensions** nikalni hogi to **Master G**

Hum formula degae

✓ Ratta nahi marna

Dimension
Formula Speed

② Velocity = $\frac{\text{displacement}}{\text{time}}$

Velocity = $L^1 T^{-1}$

$= \frac{L^1}{T^1} = L^1 T^{-1} = L T^{-1}$

$V = L T^{-1} \overset{x}{m} \overset{x}{s} \overset{x}{A} \overset{x}{K} \overset{x}{cd} \overset{x}{mol}^0$

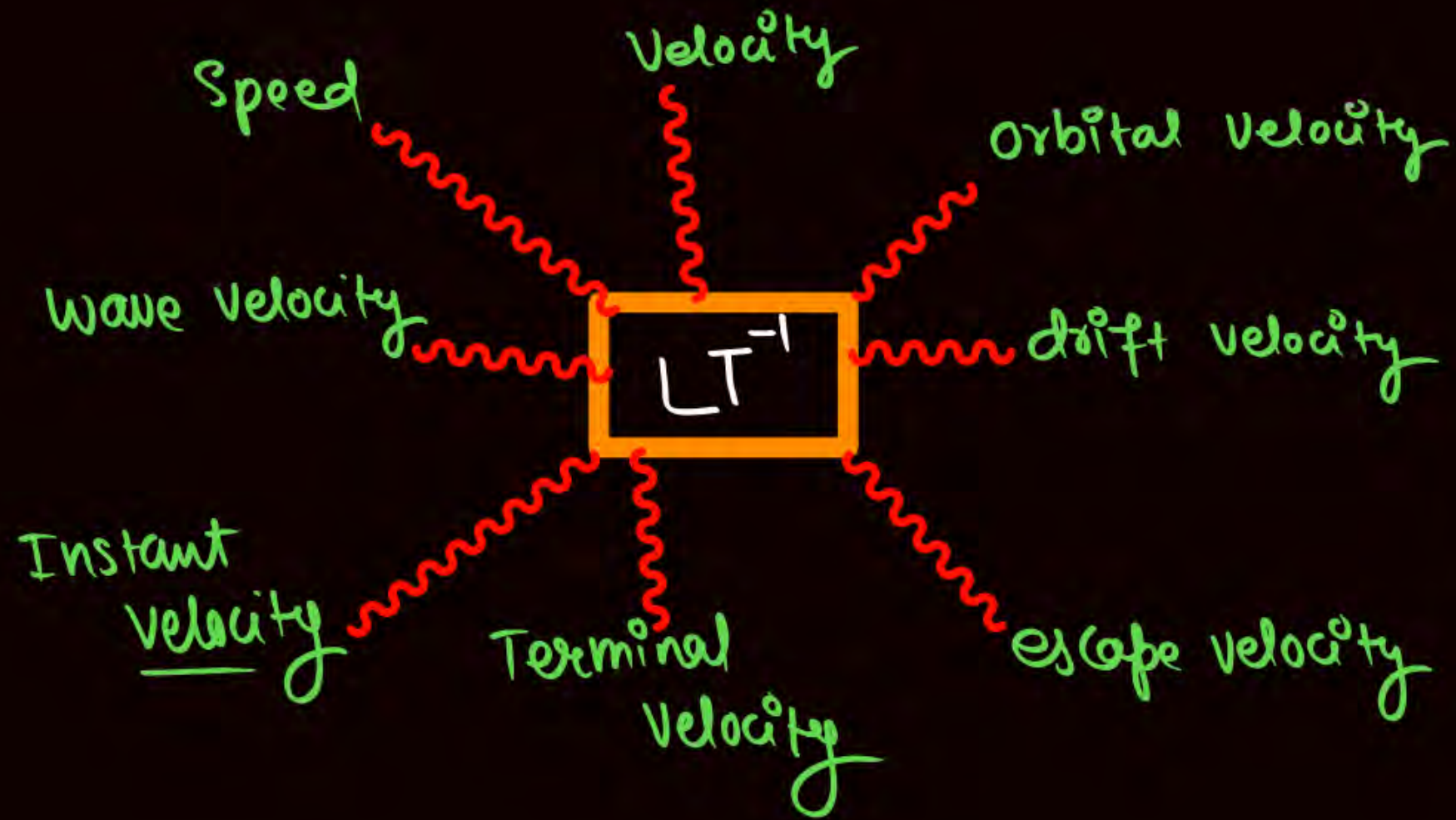
$V = L T^{-1} \times 1 \times 1 \times 1 \times 1 \times 1$

$V = L T^{-1}$

Velocity = $L^1 T^{-1} \overset{x}{m}^0$
 $= L^1 T^{-1} * *$

MLT

$\frac{L^1}{T^1}$
 $L T^{-1}$



Dothebaazz

Angular velocity

$$\begin{aligned}\omega &= \frac{\theta}{T} = \frac{1}{T'} \\ &= 1 \times T^{-1} \\ &= T^{-1}\end{aligned}$$

③ Acceleration

$$a = \frac{v}{T}$$

$$a = \frac{LT^{-1}}{T}$$

$$a = \underline{LT^{-1}} \underline{T^{-1}}$$

** $a = LT^{-2}$

$$a = m^0 (LT^{-2})$$

④ Momentum

$$p = \underline{mv} \quad \underline{LT^{-1}}$$

$$p = \underline{m} \underline{LT^{-1}}$$

$$\underline{p = MLT^{-1}}$$

⑤ Force

$$\checkmark f = \underline{ma}$$

$$F = m'LT^{-2}$$

$$\boxed{F = MLT^{-2}}$$

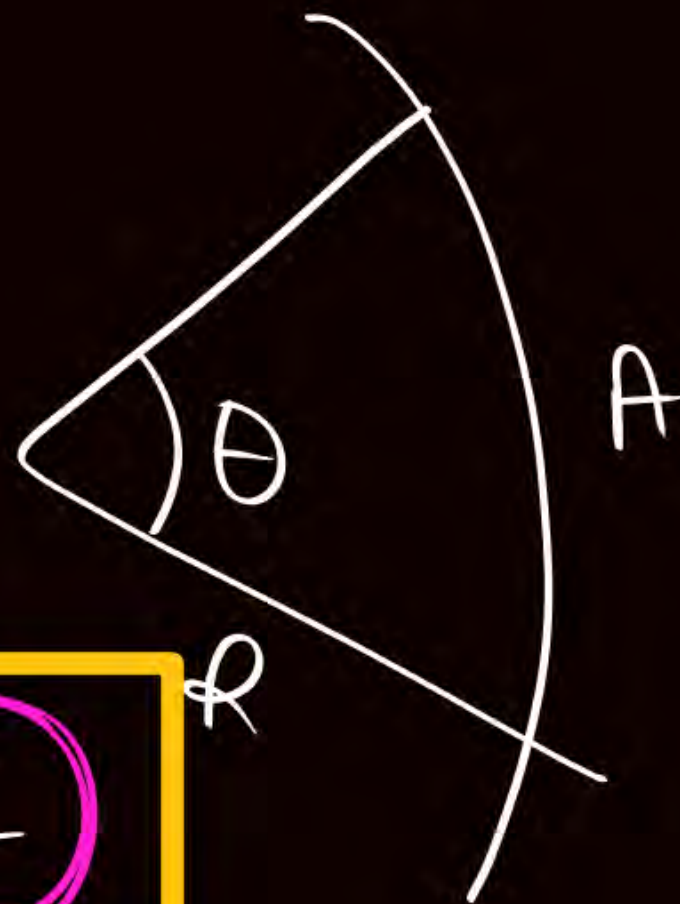
Note (g) acceleration due gravity

$$g \rightarrow a \rightarrow LT^{-2}$$

◦ Angle

$$\theta = \frac{\text{Arc}}{R}$$

$$\theta = \frac{K}{x} = 1$$



Dimensional less PQ

Lorentz Force

centripetal force

Gravitational Force

centrifugal force

friction force

Normal force

$$F = MLT^{-2}$$

Magnetic force

Electrostatic force

Tension
force

Spring force

Note

EMF

electro
motive
force

Dokhebaaz hai

$$MLT^{-2}$$

12th
↓

V

Potential difference

Surface tension
 MT^{-2}

⑥ Work $\rightarrow MLT^{-2}$

$$\underline{W} = \underline{F} \underline{x}$$

$$W = MLT^{-2} L$$

$$W = ML' \overset{\curvearrowright}{L'} T^{-2}$$

$$\underline{W = ML^2 T^{-2}}$$

⑦ Energy $\rightarrow$ Capacity to do work

$W \rightarrow E$ $\rightarrow$ Dimension K Saat

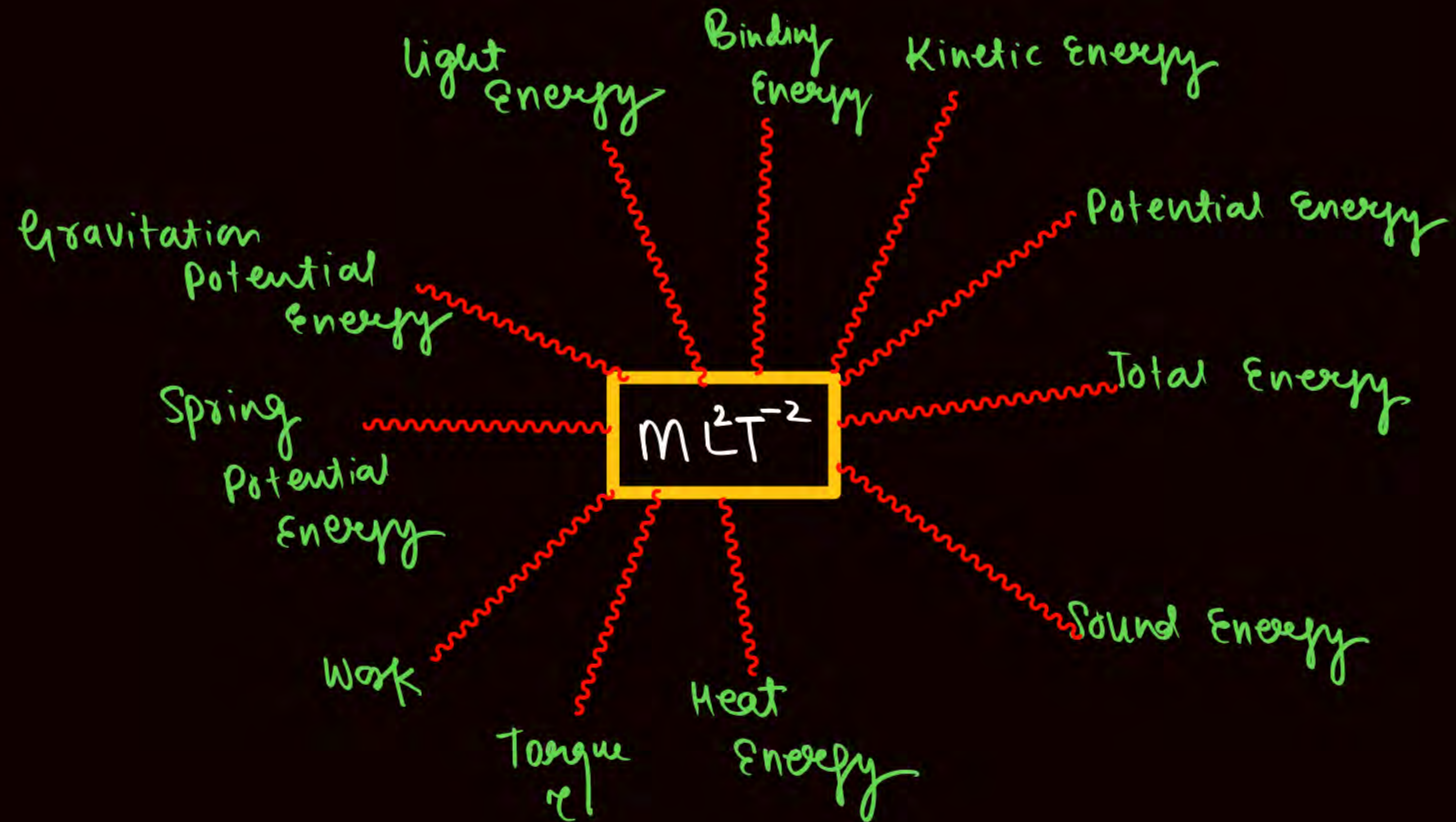
Koi lana
deng
nahi

$$\underline{K.E} = \underline{\frac{1}{2}mv^2}$$

$$\underline{K.E} = M(\underline{LT^{-1}})^2$$

$$K.E = M' L^2 T^{-2}$$

$$\underline{K.E = ML^2 T^{-2}}$$



⑧ Torque (τ)

perpendicular distance
 $\tau = \underline{r_{\perp}} \underline{F}$

$$\tau = L \underline{MLT^{-2}}$$

$$\tau = ML^2T^{-2}$$

⑨ Power

$$P = \frac{W}{T}$$

$$P = \frac{ML^2T^{-2}}{T}$$

$$P = ML^2 \underline{T^{-2}} \underline{T^{-1}}$$

$$P = ML^2T^{-3}$$

Power

⑩ Pressure / Stress
 $\text{stress} = \frac{F}{A}$

$$P = \frac{F}{A}$$

$$P = \frac{M L T^{-2}}{L^2}$$

$$P = M T^{-2} L^{-1}$$

$$\underline{P = M L^{-1} T^{-2}}$$

⑪ Area

$$A = L \times B$$

$$A = L' \times L'$$

$$A = L^2$$

⑫ Surface tension

$$ST = \frac{F}{l}$$

$$ST = \frac{M L T^{-2}}{L}$$

$$ST = M T^{-2}$$

⑬ Coefficient of friction (μ)

Force $F = \mu N$ force

✓
↓ Dimensionless po

$$F = \mu N$$

$$\frac{F}{N} = \mu$$

$$\frac{\cancel{MLT^{-2}}}{\cancel{MLT^{-2}}} = \mu$$

$$\boxed{1 = \mu}$$

Neet-2016

14) Coefficient of viscosity (η) neeta (neeta)

$$F = 6\pi \eta r v$$

$\xrightarrow{\text{Radius}}$
 $\xrightarrow{\text{velocity}}$

$$\pi = \frac{22}{7}$$

$$\frac{MLT^{-2}}{L^2T^{-1}} = \eta$$

$$\frac{MT^{-1}}{L} = \eta$$

$$MLT^{-2} = \eta L L T^{-1}$$

$$MLT^{-2} = \eta L^2 T^{-1}$$

$$\frac{MLT^{-2}}{L^2 T^{-1}} = \eta$$

$$MT^{-1}L^{-1} = \eta$$

$$M^1 L^{-1} T^{-1} = \eta$$

$$\underline{\eta = M L^{-1} T^{-1}}$$

Q (15) Gravitational Constt (Boared) (G)

$$\underline{F} = \frac{G M_1 M_2}{R^2}$$

$$G = \frac{\underline{F} \underline{R}^2}{\underline{M_1 M_2}}$$

$$F R^2 = G \underline{M_1 M_2}$$

$$G = \frac{\cancel{M} \underline{L} \underline{T}^{-2} \underline{L}^2}{\cancel{M} \underline{M'}}$$

$$\frac{F R^2}{M_1 M_2} = G$$

$$G = \bar{M}^{-1} \underline{L}^1 \underline{L}^2 \underline{T}^{-2}$$

$$G = \frac{F R^2}{M_1 M_2}$$

$$G = \bar{M}^{-1} \underline{L}^3 \underline{T}^{-2}$$

$$\boxed{G = \bar{M}^{-1} \underline{T}^{-2} \underline{L}^3}$$

QUESTIONS

What are the dimensions of coefficient of friction?

μ

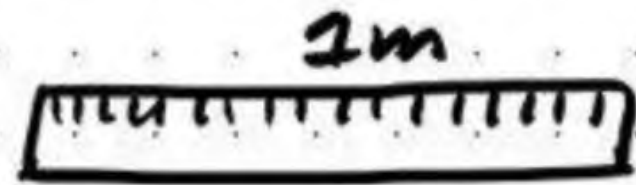
1

2

3

4

unit :- A unit is the fixed quantity in reference to which all other physical quantity are measure



1sec

System of unit

Length

MASS

Time

FPS

Foot

Pound

second.

CGS

centimetre

gram

second

MKS

metre

Kilogram

second

Dimension:- OLD units

The Dimension are the power raised to the base Quantity to determine the Relationship b/w the Derived PQ and the Base physical Quantity

Mass	M
length	L
time	T
current	A
temperature	K
luminous intensity	cd
no.-of substance	mole



Yaad
Hogaya

① $\text{Speed} = \frac{\text{distance}}{\text{time}} = \frac{L}{T} = L^1 T^{-1}$

$\text{Speed} = L^1 T^{-1}$

$\text{Speed} = 1 \times L^1 T^{-1}$

$\text{Speed} = M^0 L^1 T^{-1}$

Dimensional
Equation

Dimension

Dimensional formula

⑤ Velocity

$\text{Velocity} = \frac{\text{displacement}}{\text{time}} = \frac{L}{T} = L^1 T^{-1} = M^0 L^1 T^{-1}$

③ Acceleration

$\text{acc} = \frac{v}{T} = \frac{L^1 T^{-1}}{T} = L^1 T^{-1} T^{-1} = L^1 T^{-2} = M^0 L^1 T^{-2}$

④ Force

$F = ma = M L^1 T^{-2}$

⑤ Work

$W = Fd = M L^1 T^{-2} L = M L^2 T^{-2}$

⑥ Potential Energy

$a = g = L T^{-2}$

$P.E = mgh = M L^1 T^{-2} L = M L^2 T^{-2}$

⑦ Kinetic Energy

$K.E = \frac{1}{2} mv^2 = mv^2 = M (L T^{-1})^2 = M L^2 T^{-2}$

⑧ Power

$P = \frac{W}{T} = \frac{M L^2 T^{-2}}{T} = M L^2 T^{-2} T^{-1} = M L^2 T^{-3}$

⑨ Area

$$A = L \times B = L' \times L' = L^2$$

⑩ Volume

$$V = L \times B \times h = L' \times L' \times L' = L^3$$

⑪ Density ρ

$$\rho = \frac{M}{V} = \frac{M}{L^3} = ML^{-3}$$

⑫ Angle θ

$$\theta = \frac{\text{Arc}}{R} = \frac{L}{L} = 1 \text{ (Dimensional less Quantity)}$$

Application of Dimension-

① To find the Dimensional Formula of physical Quantity Dimensional formula of G

$$F = G \frac{M_1 M_2}{r^2}$$

$$MLT^{-2} = G \frac{MM}{L^2}$$

$$MLT^{-2} = \frac{GM^2}{L^2}$$

$$ML^2T^{-2} = GM^2$$

$$GM^2 = ML^3T^{-2}$$

$$G = \frac{ML^3T^{-2}}{M^2}$$

$$G = \frac{L^3T^{-2}}{M}$$

$$G = M^{-1}L^3T^{-2}$$

$$G = M^{-1}T^{-2}L^3$$

SUMMARY

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- Dimension

- Dimensional formula



P
A
A
η
G

Thank
You